CI360 Productivity Intelligence Suite

Complete Technical Architecture & System Documentation

Version: 2.0.0

Environment: Node.js (v18+) · Express.js · MongoDB (Mongoose) · Vite · Vanilla ES Modules

Architecture: Monorepo (Express API Backend + Vite Web Client)

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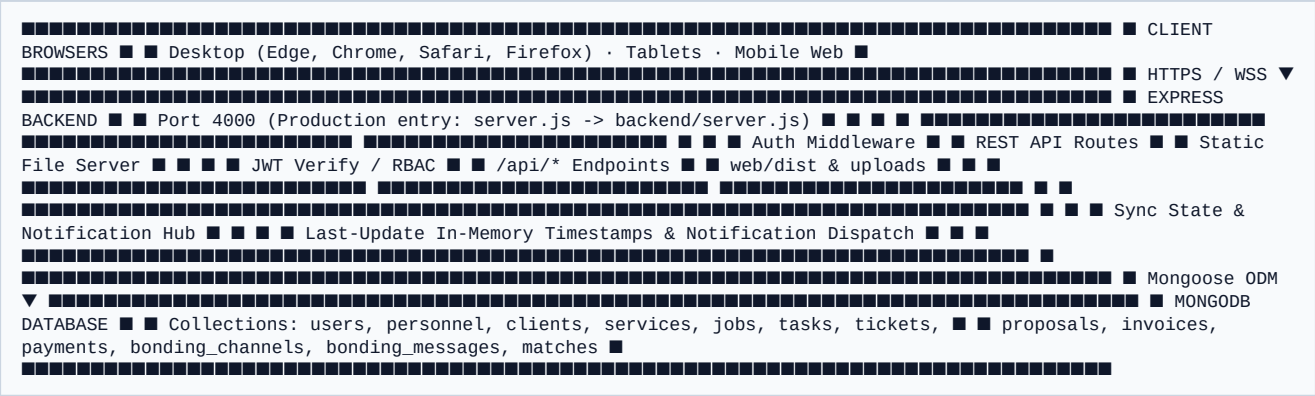
1. System Architecture Overview

CI360 operates as a high-performance, single-instance or distributed Node.js/Express service coupled with a pre-compiled Vite front-end. It provides role-based workspaces for **Super Admins**, **Admins**, **Employees**, and **Clients**.



Directory Structure

```
CIJOB/ci360/ ■■■ package.json # Root scripts (build, dev, start, seed) ■■■ server.js # Production entry point ■■■ docs/ # Technical specs and user manuals ■■■ backend/ # Node.js + Express API Server ■ ■■■ config/db.js # MongoDB connection with Mongoose ■ ■■■ middleware/auth.js # JWT verification & role authorization ■ ■■■ models/ # 21 Mongoose Data Schemas ■ ■■■ routes/ # 18 Modular Express Routers ■ ■■■ services/ # Business logic services ■ ■■■ utils/ # Notification, time & math helpers ■ ■■■ uploads/ # Ticket & invoice attachment storage ■■■ web/ # Modern Frontend (Vite + Vanilla JS + CSS) ■■■ vite.config.js # Build pipeline & development proxy ■■■ index.html # Role router landing page ■■■ login.html # Multi-tenant authentication screen ■■■ admin.html # Superadmin & Admin Workspace ■■■ employee.html # Employee Workspace ■■■ client.html # Client Extranet Portal ■■■ accounts.html # Financial Invoicing Dashboard ■■■ proposals.html # Proposal Studio Workspace ■■■ proposal.html # Client Proposal Review & Signoff ■■■ src/ ■ ■■■ css/ # style.css, tokens.css, supporttickets.css ■ ■■■ js/ # api.js, admin.js, employee.js, planner.js, ■ # supporttickets.js, proposals.js, teambonding.js ■■■ dist/ # Optimized production bundle
```

2. Database Schema & Data Models

The system maintains 21 core collections in MongoDB:

2.1 Core Identity & Operations

- **`User`**: System logins, email, bcrypt password hash, role (**`superadmin`**, **`admin`**, **`employee`**, **`client`**), status flag (**`active`**), personnel reference.
- **`Personnel`**: Staff directory, duties, employment dates, monthly hour capacity (default 160h/month), department assignment.
- **`Client`**: Client brand names, contact details, account managers, retainers, billing profiles, and project links.
- **`Service`**: Agency service offerings (Strategy, CS, Website, Design, Copy, Edit, Shoot, SEO, SMO, QC) with standard rates.
- **`Job`**: Work records tracking title, client ID, service IDs, date, completion status, value, and multi-employee percentage/hour splits.
- **`Roster`**: Staffing matrix mapping personnel to clients and ongoing assignments.
- **`SalaryGrade`** & **`SalaryAssignment`**: Confidential tier grades, pay scales, and personnel compensations (restricted strictly to Super Admin).

2.2 Task, Ticket & Project Workflow

- **`Task`**: Daily Planner tasks with title, priority (**`Low`**, **`Medium`**, **`High`**, **`Urgent`**), status (**`Pending`**, **`In Progress`**, **`Review`**, **`Completed`**), subtask checklist items, and assignees.
- **`SupportTicket`**: Scrum ticket system supporting categories (**`Bug`**, **`Feature`**, **`Task`**, **`Internal`**, **`Infrastructure`**, **`Client Request`**, **`General`**), statuses (**`To Do`**, **`Content`**, **`Creative`**, **`Under Review`**, **`Ready to Leave`**, **`In Progress`**, **`Resolved`**, **`Closed`**), priority levels, rich attachments, reporters, and linked issues (**`relates to`**, **`blocks`**, **`is blocked by`**, **`duplicates`**).
- **`Target`**: Monthly/quarterly revenue and productivity goals assigned to personnel and teams.

2.3 Sales & Billing Subsystem

- **`Proposal`**: Client proposals containing company info, scope items, deliverable milestones, pricing calculations, status (**`draft`**, **`sent`**, **`viewed`**, **`accepted`**, **`declined`**), and e-signature records.
- **`ProposalRateCard`**: Configurable service rate benchmarks.
- **`Invoice`**: Client invoices with invoice numbers, line items, CGST/SGST/IGST tax computations, payment status (**`Draft`**, **`Sent`**, **`Paid`**, **`Overdue`**, **`Cancelled`**), due dates, and PDF exports.
- **`Payment`**: Financial records tracking payment amounts, dates, bank transaction IDs, and methods.
- **`BillingProfile`**: Legal entity details, GSTIN, PAN, bank account coordinates, and payment instructions.

2.4 Team Lounge & Multiplayer Games

- **`BondingChannel`**: Communication rooms with **`type: 'direct'`** (1-on-1 private chat) or **`type: 'channel'`** (group/team lounge), membership arrays, and latest message snapshots.
- **`BondingMessage`**: Individual messages supporting text, emojis, attachments, system alerts, and interactive game challenge cards (**`type: 'challenge'`**).
- **`GameInvitation`**: Match invites specifying **`gameType`** (**`chess`**, **`tictactoe`**, **`ludo`**), sender, recipient, match settings, status (**`pending`**, **`accepted`**, **`declined`**, **`cancelled`**), and target private channel ID.
- **`GameMatch`**: Active match state storing player allocations, turn state, move logs, tournament clocks, and game-specific payloads (Chess FEN / Ludo 15×15 pawn matrices).
- **`BondingScore`**: Employee gamification scores, points awarded for challenges, wins, and daily interactions.
- **`Notification`**: System notifications for tickets, chat mentions, game challenges, and approval alerts.

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3. REST API Endpoints Reference

All API routes are prefixed with **`/api`**. Protected routes require a valid **`Bearer <token>`** HTTP header.

3.1 Authentication (`/api/auth`)

Method	Endpoint	Access	Purpose
`POST`	**`/api/auth/login`**	Public	Validates credentials; returns JWT session token (12h) & user payload
`GET`	**`/api/auth/me`**	Authenticated	Fetches current user profile and role
`POST`	**`/api/auth/change-password`**	Authenticated	Updates current user's password
`POST`	**`/api/auth/reset-password`**	Super Admin	Resets a specific employee's password

3.2 Analytics & Productivity (`/api/dashboard`, `/api/jobs`)

Method	Endpoint	Access	Purpose
`GET`	**`/api/dashboard/admin`**	Admin / Super Admin	Company-wide revenue, utilization, profit margin, capacity warnings
`GET`	**`/api/dashboard/employee`**	Employee	Personal logged hours, targets, utilization percentage
`GET`	**`/api/jobs`**	Authenticated	Lists jobs with period, client, and personnel filters
`POST`	**`/api/jobs`**	Authenticated	Logs a new job with percentage and hour splits
`PUT`	**`/api/jobs/:id`**	Authenticated	Updates existing job
`DELETE`	**`/api/jobs/:id`**	Super Admin / Creator	Deletes a job record

3.3 Daily Tasks & Support Tickets (`/api/tasks`, `/api/tickets`)

Method	Endpoint	Access	Purpose
`GET`	`/api/tasks`	Authenticated	Retrieves daily planner tasks with status filters
`POST`	`/api/tasks`	Authenticated	Creates a new task
`PUT`	`/api/tasks/:id`	Authenticated	Updates task status, checklist, or priority
`GET`	`/api/tickets`	Authenticated	Retrieves support tickets for Scrum board
`POST`	`/api/tickets`	Authenticated	Submits a new support ticket with attachments
`PUT`	`/api/tickets/:id`	Authenticated	Updates ticket status, assignees, priority, comments
`DELETE`	`/api/tickets/:id`	Creator / Superadmin	Strictly deletes a ticket (guarded by creator ID or superadmin role)

3.4 Proposals & Billing (`/api/proposals`, `/api/accounts`)

Method	Endpoint	Access	Purpose
`GET`	`/api/proposals`	Authenticated	Lists client proposals
`POST`	`/api/proposals`	Admin / Super Admin	Creates a proposal draft
`GET`	`/api/proposals/public/:token`	Public	Client access to view and sign proposal
`POST`	`/api/proposals/public/:token/sign`	Public	Records client e-signature and accepts proposal
`GET`	`/api/accounts/invoices`	Admin / Super Admin	Invoicing list with payment tracking
`POST`	`/api/accounts/invoices`	Admin / Super Admin	Generates invoice draft with tax calculations
`POST`	`/api/accounts/payments`	Admin / Super Admin	Records client payment receipts

3.5 Team Bonding & Games (`/api/bonding`)

Method	Endpoint	Access	Purpose
`GET`	`/api/bonding/channels`	Authenticated	Lists available 1-on-1 direct chats and group channels
`POST`	`/api/bonding/channels`	Authenticated	Creates group channel or direct message thread
`GET`	`/api/bonding/channels/:id/messages`	Authenticated	Loads message stream with pagination
`POST`	`/api/bonding/channels/:id/messages`	Authenticated	Posts message or media attachment
`POST`	`/api/bonding/invitations`	Authenticated	**Creates a game challenge delivered strictly to personal direct chat**
`GET`	`/api/bonding/invitations`	Authenticated	Lists pending incoming and sent challenges
`PUT`	`/api/bonding/invitations/:id/respond`	Authenticated	Accepts or declines challenge
`POST`	`/api/bonding/matches/instant-ludo`	Authenticated	Launches 2P, 3P, or 4P Ludo match
`GET`	`/api/bonding/matches/:id`	Authenticated	Fetches live game board state
`POST`	`/api/bonding/matches/:id/move`	Authenticated	Submits move or rolls 3D dice (verified turn)
`DELETE`	`/api/bonding/matches/:id`	Players / Admin	Cleans up and archives game match record

4. Authentication & Authorization Framework

4.1 JWT Token Handling

Authentication is implemented via JSON Web Tokens (JWT). Upon successful credentials validation:

1. The server signs a payload containing `{ id, role, email, name }` using `JWT_SECRET` with a 12-hour expiration.
2. The client stores this token in `localStorage.getItem('ci360_token')`.
3. Every API call attaches the header: `Authorization: Bearer <token>`.
4. Token expiration or `401 Unauthorized` responses automatically trigger redirection to `/login.html`.

4.2 Role-Based Access Control (RBAC) Matrix

Feature / Module	Super Admin	Admin	Employee	Client
Company Financial Dashboard	■ Full	■ Full	■ Blocked	■ Blocked
Personal Utilization & Jobs	■ Full	■ Full	■ Self Only	■ Blocked
Log & Edit Jobs	■ All	■ All	■ Self Only	■ Blocked
Client & Service Directory	■ Manage	■ Manage	■■ Read	■ Blocked
Salaries & Compensation	■ Full	■ Blocked	■ Blocked	■ Blocked
User Accounts & Access	■ Manage	■ Blocked	■ Blocked	■ Blocked
Support Tickets (Create/Edit)	■ Full	■ Full	■ Full	■ Blocked
Support Tickets (Delete)	■ Full	■ Blocked	■ Creator Only	■ Blocked
Proposal Studio & Signoffs	■ Full	■ Full	■■ Read	■ Client Review
Invoicing & GST Accounts	■ Full	■ Full	■ Blocked	■ Blocked
Team Chat & Games Arena	■ Full	■ Full	■ Full	■ Blocked

5. Real-time Polling & Synchronization Engine

To eliminate server overhead and maintain resilient connections across firewalls, CI360 uses an **Event-Driven Synchronization Hub**:

1. **In-Memory Timestamp Counters**:

The backend tracks global modification timestamps:

- `__ci360LastGlobalUpdate`
- `__ci360LastTicketUpdate`
- `__ci360LastTaskUpdate`
- `__ci360LastInvoiceUpdate`
- `__ci360LastChatUpdate`
- `__ci360LastBondingUpdate`

2. **Lightweight Heartbeat Polling**:

Clients ping `/api/notifications/sync-status` every 3 seconds:

- If timestamps match the client's cached state, the server returns `{ modified: false }` with zero database overhead.
- If an entity was updated (e.g., ticket moved, dice rolled, chat sent), the client re-fetches only the affected view without re-rendering the whole page.

6. Design System, Theming & Layout Spacing Standard

6.1 Theme Tokens & Dark Mode

CI360 supports **Light Theme** (default) and **Dark Theme** via CSS custom variables (`[data-theme="dark"]`). Tokens adapt typography, borders, surfaces, and shadows seamlessly across all screens.

6.2 Global Content Spacing Standard

All primary views inside the application shell are governed by an explicit CSS container standard in `style.css`:

```
.app-content { max-width: none !important; width: auto !important; box-sizing: border-box !important; flex: 1 1 0%;
padding: 20px !important; margin: 0px 71px !important; } @media (max-width: 992px) { .app-content { margin: 0 32px
!important; padding: 20px !important; } } @media (max-width: 768px) { .app-content { margin: 0 16px !important;
padding: 16px !important; } }
```

- **Uniform 20px Padding**: Applied inside the main viewport on top, right, bottom, and left.
- **71px Horizontal Desktop Margin**: Establishes a comfortable reading margin on widescreen desktop displays.
- **Responsive Fallback**: Gracefully scales to 32px on tablets and 16px on mobile viewports.

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7. Subsystem Architecture

7.1 Executive Dashboard & Capacity Intelligence

Aggregates billable hours against employee capacity (default 160 hrs/mo) and calculates workload utilization percentages:

$$\text{Utilization \%} = \left(\frac{\text{Logged Billable Hours}}{\text{Monthly Target Hours}}\right) \times 100$$

- Highlights over-utilized (>100%) and under-utilized (<70%) team members with proactive capacity redistribution advice.

7.2 Job Logging & Billable Split Engine

Supports multi-person work distribution on a single job:

- Multiple team members can be assigned percentage splits $\sum \% = 100\%$ or individual logged hours.
- Revenue recognition automatically divides client billings across participating personnel.

7.3 Daily Planner & Task Kanban

Features a priority matrix (Urgent, High, Medium, Low), inline task creation, subtask checklist completion percentages, and drag-and-drop or click-to-move status updates.

7.4 Support Tickets & Jira Scrum System

- 5-Column to 8-Column Scrum board with drag/select progression.
- Fast streaming multi-file upload system and external reference URL linking.
- **Strict Security Guard**: Only the original ticket reporter or a Superadmin may delete a ticket.

7.5 Proposal Studio & Rate Card Engine

Interactive proposal generator:

- Scopes deliverables, milestones, payment schedules, and terms.
- Generates secure public client links for digital signature and acceptance.

7.6 Accounts, Invoicing & GST Financials

- Automatically formats invoices compliant with Indian GST (CGST 9% + SGST 9% or IGST 18%).

- Tracks accounts receivable, aging buckets (0-30 days, 31-60 days, 60+ days), and payment logs.

7.7 Team Chat, Lounge & Multiplayer Games Arena

- **Personal Chat Challenge Delivery**: Every challenge generates an invitation strictly into the 1-on-1 direct conversation between the two colleagues.
- **Ludo Royale**: 15×15 grid cross track, 4 pawns per color, auto-advance for single valid moves, safe squares, and a 3D tactile dice with realistic animations and isolated styling.
- **Chess Duel**: Standard FEN validation via `chess.js`, legal move dot indicators, live clocks, and notation.
- **Tic-Tac-Toe**: Turn-based fast match.

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8. Production Build, Deployment & DevOps

8.1 Build Pipeline

The web client is compiled using Vite:

```
cd web npm run build
```

This builds production bundles into `web/dist/`, minifying JavaScript, optimizing stylesheets, and tree-shaking unused libraries.

8.2 Production Server Execution

The Express application serves the API and the compiled frontend simultaneously:

```
# In project root npm run start
```

Runs `node server.js` which loads environment variables and serves `web/dist` on port 4000.